
DATP

Research Dossier

Device-Aware Threshold Personalization for Non-IID Federated IoT Anomaly Detection

Version: v1.0

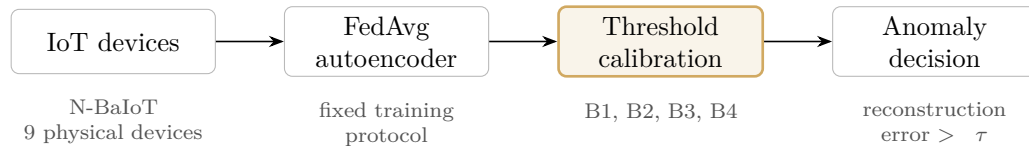
Status: Scientific and technical scoping document

Scope: N-BaIoT threshold-calibration study. The autoencoder architecture, the FedAvg training protocol, and the trained score artifacts are fixed across B1–B4; only threshold computation scope changes.

Purpose

- Study threshold calibration scope in federated IoT anomaly detection.
- Isolate the threshold policy as the changing variable.
- Keep the FedAvg autoencoder and score artifacts fixed across B1–B4.
- Measure per-client false-positive-rate disparity.
- Compare shared, per-client, family-mean, and cluster-mean threshold policies.
- Bound claims to the tested datasets, partitions, seeds, and protocol.

Minimal Pipeline



Only the *threshold calibration* step is varied across the controlled ladder.

Scope Boundary

- Threshold policy may change.
- Model architecture does not change.
- FedAvg protocol does not change.
- Score artifacts are reused across policies.
- B0 is a centralized reference only.
- Privacy, deployment, poisoning, and model personalization are out of scope.

Project Identity

Field	Value
Project acronym	DATP
Full title	Device-Aware Threshold Personalization for Non-IID Federated IoT Anomaly Detection
Document type	Research Dossier
Version	v1.0
Status	Scientific and technical scoping document
Thesis line	Collaborative Malware Detection with Federated Learning in IoT Networks
Research branch	Threshold personalization / performance-equity branch
Main dataset	N-BaIoT
Secondary dataset	CICIoT2023 file-level applicability check
Client definition	Physical IoT device as FL client for Regime A
Model setting	Fixed FedAvg autoencoder
Threshold policies	B1, B2, B3, B4
Primary metric	CV(FPR) across eligible clients
Main claim scope	Regime A, B1 vs. B2, threshold-scope effect

Document Purpose

This document defines:

- Research scope.
- Scientific objectives.
- Methodological boundaries.
- Validation criteria.
- Expected deliverables.
- Risks and limitations.

This document is not:

- A paper draft.
- A presentation deck.
- An implementation log.
- A roadmap or internal planning document.

Context and Positioning

Problem Context

- Federated IoT anomaly detection often trains a shared autoencoder.
- Detection requires a reconstruction-error threshold.
- Heterogeneous devices can have different benign reconstruction-error distributions.
- A shared threshold can impose unequal false-positive rates across clients.
- DATP studies threshold scope as the operating-point variable.

DATP Rationale

- DATP fixes the model and training protocol.
- DATP varies only threshold calibration scope.
- This allows attribution to threshold policy rather than architecture or aggregation.
- The central question is whether local or grouped thresholds reduce per-client FPR disparity.

Three-Branch Comparison

	DATP	DATP-CP	Journal extension
Scientific role	Threshold personalization and FPR-equity analysis.	Calibration-channel vulnerability analysis.	External validation and generalization.
Main question	How much does threshold scope reduce per-client FPR disparity?	Can calibration contamination shift thresholds and harm detection?	Does the DATP threshold-scope effect survive a stronger protocol?
Primary variable	Threshold policy scope.	Calibration contamination.	Datasets, comparators, threshold variants, stress tests.
Adversary	None.	Single compromised client / gray-box score access.	None.
Claim boundary	Tested datasets and partitions; primary Regime A B1 vs. B2.	N-BaIoT calibration-channel poisoning.	External datasets.

Key Distinctions

- **DATP:** performance and FPR-equity branch.
- **DATP-CP:** security and calibration-channel poisoning branch.
- **Journal extension:** validation and generalization branch.

Problem

- Thresholds are computed after federated training.
- Thresholds define the final anomaly decision boundary.
- Benign calibration scores determine τ .
- Under non-IID device heterogeneity, one shared τ may not fit all clients.
- Per-client FPR disparity is operationally important because false alarms are experienced locally.

Central Research Question

How much does threshold calibration scope reduce per-client false-positive-rate disparity under a fixed FedAvg autoencoder in non-IID federated IoT anomaly detection?

Research Questions

- **RQ1:** How large is per-client FPR disparity under the client-averaged shared threshold?
- **RQ2:** Does per-client threshold calibration reduce CV(FPR) relative to the client-averaged shared threshold under Regime A?
- **RQ3:** Can cluster-mean calibration approximate per-client calibration without a device taxonomy?

Objectives

Scientific objectives

- Isolate threshold calibration scope as the experimental variable.
- Quantify per-client FPR dispersion under B1.
- Measure B1-to-B2 CV(FPR) reduction.
- Evaluate B4 as a taxonomy-free approximation.
- Characterize detection-quality tradeoffs.

Technical objectives

- Preserve fixed FedAvg autoencoder training.
- Reuse the same score artifacts across threshold policies.
- Apply eligibility rules consistently.
- Report CV(FPR), CV(TPR), worst-client BA, Macro-F1, and P10 Macro-F1.
- Preserve reproducibility of tables, figures, and seeds.

Scope

Included	Excluded
N-BaIoT Regime A as primary study	Training-data poisoning
Physical device clients	Model poisoning
Fixed FedAvg autoencoder	Aggregation poisoning
B1 client-averaged shared threshold	Backdoors
B2 per-client threshold	Evasion
B3 family-mean threshold, Regime A only	Privacy mechanisms
B4 cluster-mean threshold	Deployment benchmarking
CICIoT2023 file-level applicability boundary	Hardware validation
Regime C Dirichlet severity sweep	Concept drift
CV(FPR) over eligible clients	Model-personalization comparators
B0 centralized reference comparator	Universal IoT malware detection claims
	Formal privacy guarantees

Methodology and Validation

Experimental Setup

Dataset — N-BaIoT, 9 physical IoT devices; one physical device equals one FL client in Regime A. CICIoT2023 is a file-level applicability check, not physical-device evidence. Regime C uses Dirichlet repartitioning of N-BaIoT.

Model — Federated autoencoder; FedAvg; fixed architecture and training protocol; B1–B4 share the same trained model and score artifacts.

Calibration — Benign reconstruction-error scores; $q = 0.95$ / p_{95} threshold rule; eligible clients require sufficient benign calibration support; calibration-pending clients use the shared fallback threshold where applicable.

Threshold Policies

- **B1:** client-averaged shared threshold.
- **B2:** per-client threshold.
- **B3:** family-mean threshold, Regime A only.
- **B4:** cluster-mean threshold using 4-scalar benign-error fingerprints.

Validation Criteria

Criterion	Measurement	Interpretation
B1 dispersion	CV(FPR) under B1	Shared-threshold disparity
B1–B2 reduction	$\Delta = \text{CV(FPR)}_{B1} - \text{CV(FPR)}_{B2}$	Main threshold-scope effect
B4 recovery	$(B1 - B4)/(B1 - B2)$	Taxonomy-free approximation
Detection tradeoff	CV(TPR), Macro-F1, P10 Macro-F1	Whether FPR equity affects detection quality
Applicability boundary	CICIoT2023 Regime B	Whether effect appears under near-homogeneous file-level partition
Severity behavior	Regime C α sweep	Whether effect scales with heterogeneity
Coverage	eligible / total clients	Whether dispersion metrics are well supported

Main results (Regime A, N-BaIoT, 5 seeds). B1 CV(FPR) = 1.017; B2 CV(FPR) = 0.299; $\Delta_{B1-B2} = 0.718$, all five seed-level deltas positive. B4 recovers approximately 52% of B2’s improvement. Under the tested CICIoT2023 file-level partition (Regime B), no personalization improvement is observed.

Claim Gates

- **Gate 1:** B1 exhibits operationally relevant FPR dispersion.
- **Gate 2:** B2 reduces CV(FPR) relative to B1 in Regime A.
- **Gate 3:** detection-quality tradeoffs are reported, not hidden.
- **Gate 4:** secondary regimes are not elevated to the main claim.

Deliverables

- D1** Research protocol and scope definition.
- D2** Threshold-policy implementation.
- D3** Clean experiment outputs.
- D4** Figures and result tables.
- D5** Conference manuscript.
- D6** Reproducibility package.
- D7** Presentation dossier.
- D8** Journal-extension handoff notes, as future branch positioning.

Risks and Limitations

- N-BaIoT has only 9 physical devices.
- B4 is exploratory at small K .
- CICIoT2023 file-level clients are not physical-device clients.
- $E = 1$ limits local drift.
- Five seeds limit statistical power.
- No adversarial robustness evaluation.
- No privacy mechanism.
- No hardware or deployment validation.
- No concept drift modeling.

Work Packages

- WP1** Scope and claim-boundary audit.
- WP2** Source-material extraction and verification.
- WP3** LaTeX dossier creation.
- WP4** Scientific consistency audit.
- WP5** Visual/render audit.
- WP6** Final package cleanup.

Governance

- Keep the main claim centered on Regime A B1 vs. B2 CV(FPR).
- Keep B0 outside the controlled threshold ladder.
- Keep B3 Regime A only.
- Treat Regime B as an applicability boundary.
- Treat Regime C as secondary severity evidence.
- Do not claim universal personalization benefit.
- Do not claim privacy or deployment readiness.
- Do not mix DATP with DATP-CP poisoning claims.
- Do not absorb journal-extension goals.

Research Contribution

DATP contributes:

- A threshold-scope controlled comparison.
- Per-client FPR-dispersion evidence under physical-device heterogeneity.
- A per-client threshold personalization baseline.
- A taxonomy-free cluster approximation.
- A bounded empirical foundation for the later DATP-CP security branch and journal extension.